Statistical Inference
(통계적 추론)

Kyungtaek Park, Ph.D.

Department of Statistics
Jeonbuk National University

Today's Topics

Statistical Inference

- Probability Distribution
- Statistical Inference for a Proportion
- Statistical Inference for Discrete Data

Probability Distribution

Probability distribution functions for different data structure

- T-test, ANOVA, Regression analysis: Normal
- Survival analysis: Exponential, Weibull
- Categorical data analysis: Poisson, binomial, multinomial
- Generalized Linear Model: exponential family (normal, Poisson, binomial, multinomial, etc)
- Nonparameteric statistics: do not assume any specific model

Probability Distribution (Cont'd)

Poisson Distribution

Represents the distribution of the number of events occurring within a fixed time or space interval

(Example: The number of traffic accidents occurring within a specific time period or region)

Probability mass function (PMF)

$$p(y) = P(Y = y) = \frac{e^{-\lambda}\lambda^{y}}{y!},\ y = 0, 1, 2, \cdots .$$

$$E(Y) = \lambda,\ var(Y) = \lambda$$

Because the variance increases with the mean, the observed counts become more variable as the average count grows.

Over-dispersion: A phenomenon where the observed variance exceeds the mean, violating the basic Poisson assumption

Probability Distribution (Cont'd)

Poisson Distribution

Additivity property

$$Y_1 \sim P(\lambda_1) \perp Y_2 \sim P(\lambda_2) \rightarrow Y_1 + Y_2 \sim P(\lambda_1 + \lambda_2)$$

Probability Distribution (Cont'd)

Conditional Distribution of Poisson Variables

Suppose $Y_1 \sim P(\lambda_1) \perp Y_2 \sim P(\lambda_2)$

$$P(Y_1 = y | Y_1 + Y_2 = n) = \frac{P(Y_1 = y, Y_2 = n - y)}{P(Y_1 + Y_2 = n)}$$

$$= \binom{n}{y} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2} \right)^y \left(\frac{\lambda_2}{\lambda_1 + \lambda_2} \right)^{n-y}$$

(Example) In a certain area, the average number of injury accidents is 1, and the average number of non-injury accidents is 3. If a total of 10 accidents occurred, what is the probability that 2 of them were injury accidents?

Probability Distribution (Cont'd)

Bernoulli Distribution

Represents an experiment with only two possible outcomes, such as success/failure.

Let the random variable Y be defined as

$$Y = \begin{cases} 0, & \text{failure} \\ 1, & \text{success} \end{cases}$$

and suppose $P(success) = \pi$, then

$$p(y) = P(Y = y) = \pi^y (1 - \pi)^{1-y},\ y = 0, 1$$

$$E(Y) = \pi,\ \sigma^2(Y) = \pi(1 - \pi)$$

Probability Distribution (Cont'd)

Binomial Distribution

Represents the distribution of the number of successes in n independent Bernoulli trials, each with the same probability of success π.

In other words, it is the sum of independent Bernoulli random variables:

$$\begin{array}{rcccccccc} & Y_1 & + & Y_2 & +\cdots+ & Y_n & = & Y \\ \text{success} & 1 & & 1 & \cdots & 1 & & \downarrow \\ \text{failure} & 0 & & 0 & \cdots & 0 & & \text{\# of success} \end{array}$$

PMF

$$p(y) = P(Y = y) = \binom{n}{y} \pi^y (1-\pi)^{n-y},\ y = 0, 1, \cdots, n.$$

$$E(Y) = n\pi,\ \sigma^2(Y) = n\pi(1-\pi).$$

The variance of the binomial distribution is smaller than its mean.

Over-dispersion: Occurs when the observed variance of Y is greater than $n\pi(1-\pi)$.

Probability Distribution (Cont'd)

Poisson Approximation of the Binomial Distribution

When the number of trials n is large and the success probability π is small, the binomial distribution can be approximated by a Poisson distribution.

This approximation is generally valid when $n\pi \leq 5$.

Let $n\pi = \mu$, then $Y \sim P(\mu)$

(Example) Suppose, on average, there is 1 bug per 500 lines of code. If a programmer writes five programs, each with 300 lines, what is the probability that 2 or fewer bugs occur in total?

Probability Distribution (Cont'd)

Normal Approximation of the Binomial Distribution

When both $n\pi > 5$ and $n(1-\pi) > 5$, the central limit theorem allows the binomial distribution to be approximated by a normal distribution:

$$Y \sim N(n\pi, n\pi(1-\pi))$$

When the number of trials n is large and the success probability $\pi = 0.5$, the binomial distribution is symmetric; the closer π is to 0.5, the faster the normal approximation converges.

y	π=0.2	π=0.5	π=0.8
0	0.107	0.001	0.000
1	0.268	0.010	0.000
2	0.302	0.044	0.000
3	0.201	0.117	0.001
4	0.088	0.205	0.005
5	0.027	0.246	0.027
6	0.005	0.205	0.088
7	0.001	0.117	0.210
8	0.000	0.044	0.302
9	0.000	0.010	0.268
10	0.000	0.001	0.107

Probability Distribution (Cont'd)

Multinomial Distribution

The experiment has c possible outcomes.

Let the probability of each outcome be π_i for $i = 1,2,\dots,c$, where $\sum_{i=1}^{c} \pi_i = 1$

When the experiment is repeated n times independently, the joint distribution of the counts for each outcome follows the multinomial distribution:

$$p(y_1, \cdots, y_{c-1}) = \frac{n!}{y_1! \cdots y_c!} \pi_1^{y_1} \cdots \pi_c^{y_c},\ y_i \geq 0,\ \sum_{i=1}^{c} y_i = n$$

$$E(Y_i) = n\pi_i,\ \sigma^2(Y_i) = n\pi_i(1-\pi_i),\ \sigma(Y_i, Y_j) = -n\pi_i\pi_j$$

(Example) Suppose $X_1 \sim$ Poisson$(\lambda_1), \dots, X_k \sim$ Poisson(λ_k) are independent random variables. Then, show the conditional distribution follows a multinomial distribution:

$$P(Y_1 = y_1, Y_2 = y_2 | Y_1 + Y_2 + \dots + Y_k = n)$$

Statistical Inference for a Proportion

Likelihood function

Suppose a random variable Y follows a probability density function $p_Y(y \mid \theta)$ with parameter θ.

Then, the likelihood function of θ is defined as:

$$L(\theta) \equiv p_Y(y|\theta)$$

The density function is a function of the variable y, while the likelihood function is a function of the parameter θ.

The log-likelihood function is given by $l(\theta) \equiv \log L(\theta)$

(Example) Let $X \sim U(0, \theta)$, then draw then the likelihood function of θ is when the observed value of X is x.

Statistical Inference for a Proportion (Cont'd)

Maximum Likelihood Estimation

Given observed data, the maximum likelihood estimation method selects the parameter value θ that maximizes the likelihood function $L(\theta)$.

In general, the MLE $\hat{\theta}$ is obtained by differentiating the likelihood function with respect to θ and setting the derivative equal to zero

It is often more convenient to maximize the log-likelihood function

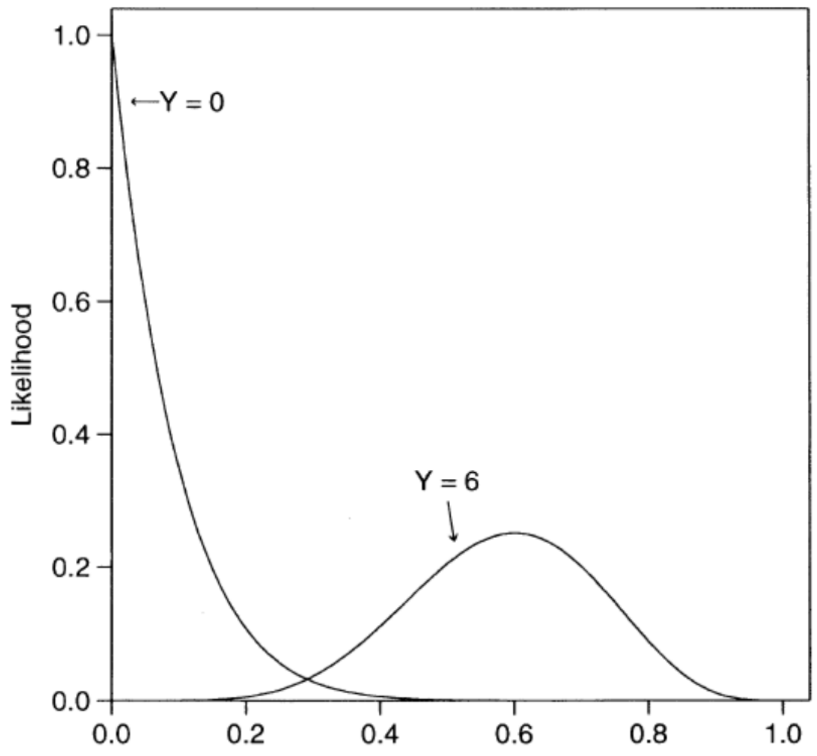
If the MLE of the parameter θ is unique (denoted $\hat{\theta}$), then the MLE of any function of the parameter $g(\theta)$ is given by:

$$\hat{g} = g(\hat{\theta})$$

That is, $g(\hat{\theta})$ is the MLE of $g(\theta)$.

Statistical Inference for a Proportion (Cont'd)

Maximum Likelihood Estimation



<Likelihood function of binomial distribution ($n = 10$)>

Statistical Inference for a Proportion (Cont'd)

Maximum Likelihood Estimation

(Example) Suppose $Y \sim \mathrm{B}(n,\pi)$,where n takes values 2 or 3, and π takes values $\frac{1}{2}$ or $\frac{1}{3}$. In this case, for each combination of n and π, determine the maximum likelihood estimate

(Example) If $Y \sim B(n,\pi)$ and we observe $Y = y$ with known n, find the MLE of π.

(Example) If $Y_1, \dots, Y_n \overset{\text{i.i.d.}}{\sim} N(\mu, 1)$, find the MLE of μ.

(Example) If $Y_1, \dots, Y_n \overset{\text{i.i.d.}}{\sim} \mathrm{Poisson}(\lambda)$:

(1) Write the log-likelihood for λ.

(2) Find the MLE of λ.

(3) Find the MLE of $P(Y_1 = 0)$.

Statistical Inference for a Proportion (Cont'd)

Significance Test for a Binomial Proportion

Null hypothesis: $H_0 : \pi = \pi_0$

Alternative hypothesis: $H_1 : \begin{cases} \pi > \pi_0 \\ \pi < \pi_0 \\ \pi \neq \pi_0 \end{cases}$

When the sample size n is large:

Score test statistic: $Z = \dfrac{p - \pi_0}{\sqrt{\dfrac{\pi_0(1-\pi_0)}{n}}} \sim N(0,1)$, where $p = Y/n$.

Wald test statistic: $Z = \dfrac{p - \pi_0}{\sqrt{\dfrac{p(1-p)}{n}}} \sim N(0,1)$

P-value: $p = \begin{cases} P(Z \geq z) \\ P(Z \leq z) \\ 2 \times P(Z \geq |z|) \end{cases}$

Statistical Inference for a Proportion (Cont'd)

Significance Test for a Binomial Proportion

(Example) A survey was conducted on whether to legalize abortion for pregnant women.

$$H_0: \pi = 0.5 \text{ vs } H_1: \pi < 0.5$$

At the University of Chicago, among 893 respondents, 400 were in favor and 493 were opposed. What conclusion can be drawn?

Statistical Inference for a Proportion (Cont'd)

Significance Test for a Binomial Proportion

When the sample size n is small:

P-value:

$$p = \begin{cases} P(Y \geq y) \\ P(Y \leq y) \\ \sum_{t \in T_y} P(Y = t) \end{cases} \quad \text{where } T_y = \{t : P(Y=t) \leq P(Y=y)\}$$

Mid *P*-value:

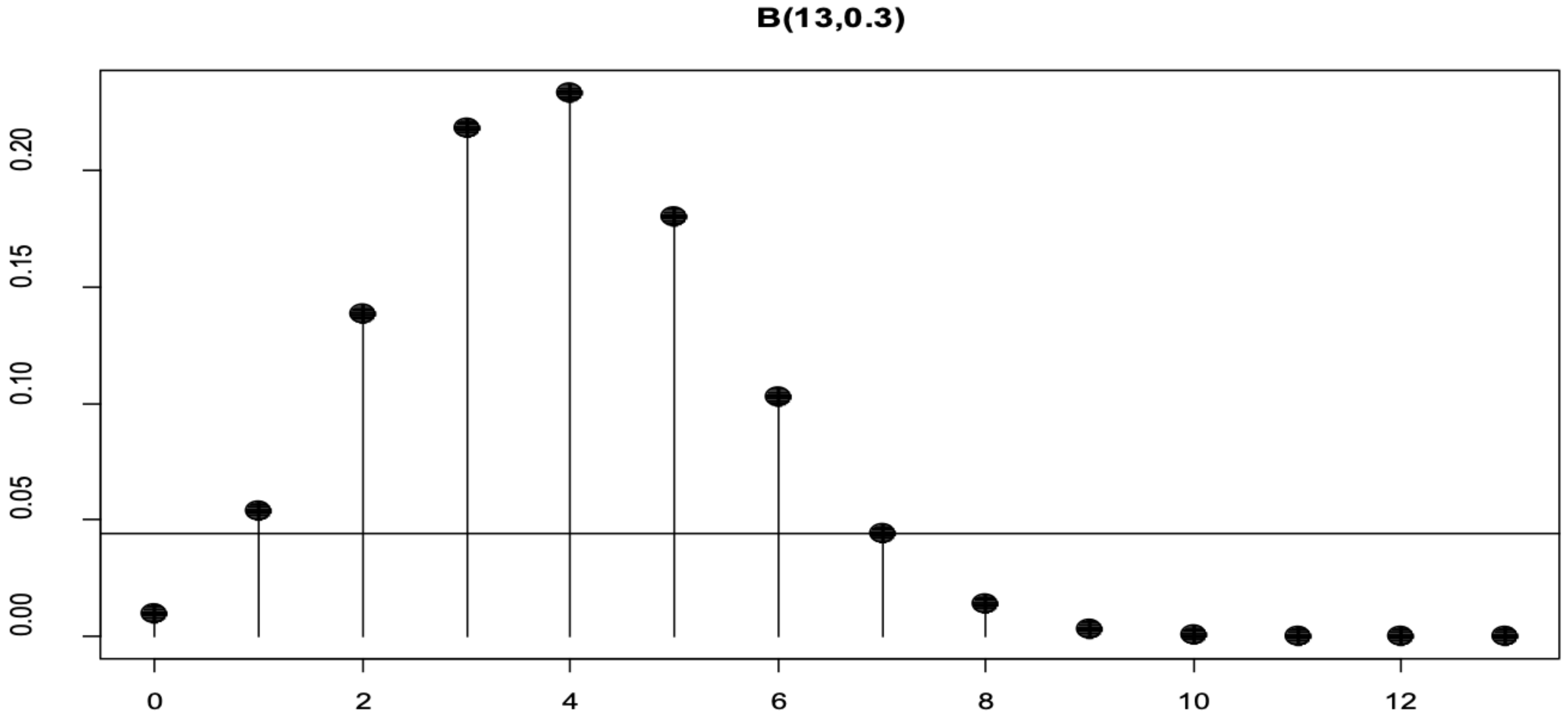
$$p = \begin{cases} 0.5P(Y=y) + P(Y>y) \\ 0.5P(Y=y) + P(Y<y) \\ 0.5\sum_{t \in T_y^1} P(Y=t) + \sum_{t \in T_y^2} P(Y=t) \end{cases}$$

$$\text{where } T_y^1 = \{t : P(Y=t) = P(Y=y)\},\ T_y^2 = \{t : P(Y=t) < P(Y=y)\}$$

Statistical Inference for a Proportion (Cont'd)

Significance Test for a Binomial Proportion

(Example) The previous highest cure rate is 0.3. A new anticancer drug was tested on $n = 13$ patients, and $y = 7$ were cured (≥5-year survival). Test whether the new drug is more effective than the old one. Compute both the p-value and the mid-p value.



Statistical Inference for a Proportion (Cont'd)

Significance Test for a Binomial Proportion

Under a true null, the p-value should follow Uniform(0,1); when n is small, the p-value follows discrete distribution and $E[p] > 0.5$.

(Example) Assume $Y \sim B(10, \pi)$ .Test H_0: $\pi = 0.5$ vs H_1: $\pi > 0.5$ $(n = 10)$. For an observed count y, define expected values of p-value and mid p-value

y	$P(y)$	P-value	Mid P-value
0	0.001	1.000	0.9995
1	0.010	0.999	0.994
2	0.044	0.989	0.967
3	0.117	0.945	0.887
4	0.205	0.828	0.726
5	0.246	0.623	0.500
6	0.205	0.377	0.274
7	0.117	0.172	0.113
8	0.044	0.055	0.033
9	0.010	0.011	0.006
10	0.001	0.001	0.0005

Statistical Inference for a Proportion (Cont'd)

Confidence Interval for a Binomial Proportion

When the number of trials n is large, the Central Limit Theorem implies:

$$p = \frac{Y}{n} \sim N\left(\pi, \frac{\pi(1-\pi)}{n}\right) \rightarrow Z = \frac{p-\pi}{\sqrt{\frac{\pi(1-\pi)}{n}}} \sim N(0,1)$$

The 100(1−α)% confidence interval for π:

Wald confidence interval

$$p - z_{\alpha/2}\sqrt{\frac{p(1-p)}{n}}, p + z_{\alpha/2}\sqrt{\frac{p(1-p)}{n}}$$

Statistical Inference for a Proportion (Cont'd)

Confidence Interval for a Binomial Proportion

The 100(1−α)% confidence interval for π:

Score confidence interval

$$\left[\frac{p+\frac{Z_{\alpha/2}^2}{2n}-Z_{1-\alpha/2}\sqrt{\frac{p(1-p)}{n}+\frac{Z_{\alpha/2}^2}{4n^2}}}{1+\frac{Z_{\alpha/2}^2}{n}},\frac{p+\frac{Z_{\alpha/2}^2}{2n}+Z_{1-\alpha/2}\sqrt{\frac{p(1-p)}{n}+\frac{Z_{\alpha/2}^2}{4n^2}}}{1+\frac{Z_{\alpha/2}^2}{n}}\right]$$

Adjusted Wald 95% confidence interval

$$\left[\tilde{\pi}-Z_{1-\alpha/2}\sqrt{\frac{\tilde{\pi}(1-\tilde{\pi})}{n}},\tilde{\pi}+Z_{1-\alpha/2}\sqrt{\frac{\tilde{\pi}(1-\tilde{\pi})}{n}}\right] \text{ where } \tilde{\pi}=\frac{np+2}{n+4}.$$

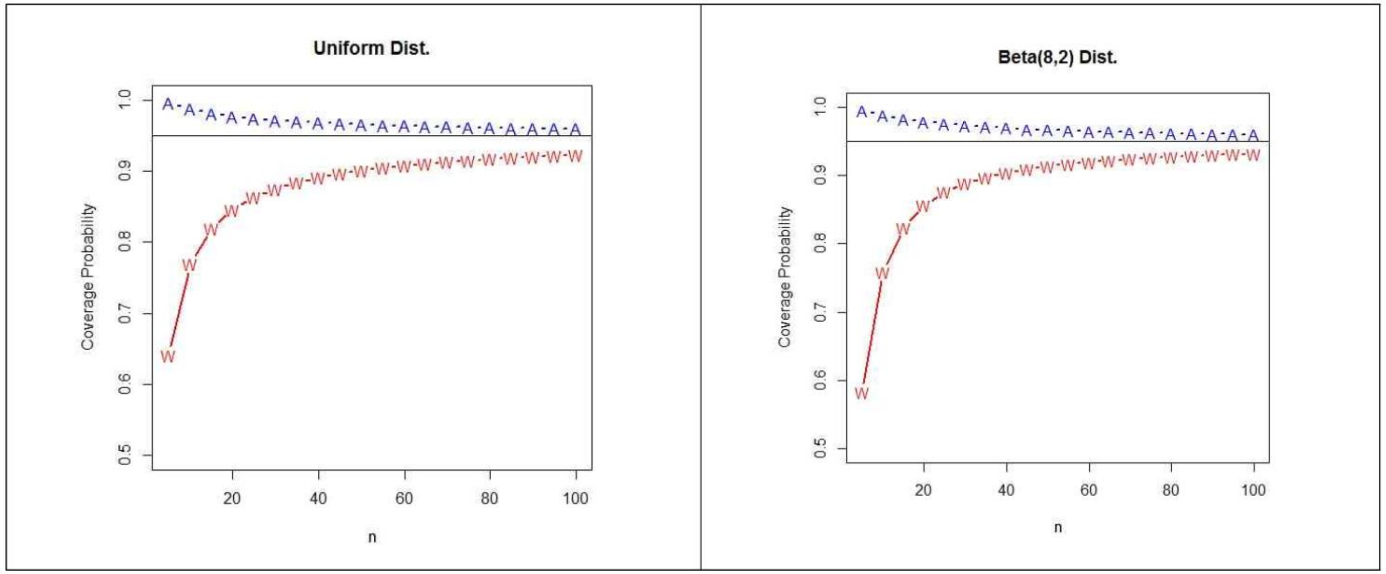
Statistical Inference for a Proportion (Cont'd)

Discussion

The Wald confidence interval performs poorly when the sample size n is not large or when the observed proportion p is close to 0 or 1.

$$P(L \le \pi \le U) = 1 - \alpha$$

Simulation results at the nominal 95% confidence interval



<A: adjusted Wald CI vs W: Wald CI>

Statistical Inference for a Proportion (Cont'd)

(Example) In a public opinion survey evaluating the local government system, out of 500 respondents, 165 gave a positive response. Construct a 95% confidence interval for the proportion of people in the entire population who have a positive opinion about the local government system.

$$p = \frac{165}{500} = 0.33,\ s.e.(p) = 0.021$$

Statistical Inference for Discrete Data

Definition

$$L_n(\pi) \equiv f_{Y_1, \dots, Y_n}(y_1, \dots, y_n)$$

$$l_n(\pi) \equiv \log L_n(\theta)$$

Score: $\dfrac{\partial l_n(\theta)}{\partial \theta}$

If $Y_1, \dots, Y_n$ are independent, $\dfrac{\partial l_n(\theta)}{\partial \theta} = \sum_{i=1}^{n} \dfrac{\partial \log f_{Y_i}(y_i;\theta)}{\partial \theta}$

$$I_n(\theta) \equiv E\left\{\left(\frac{\partial l_n(\theta)}{\partial \theta}\right)^2\right\}, \quad I_1(\theta) \equiv E\left\{\left(\frac{\partial \log f_{Y_1}(Y_1;\theta)}{\partial \theta}\right)^2\right\}$$

Statistical Inference for Discrete Data (Cont'd)

Property

When the regularity conditions (R0)–(R4) are satisfied,

$$E\left\{\left(\frac{\partial l_n(\theta)}{\partial\theta}\right)\right\} = 0,\ I_n(\theta) = -E\left\{\left(\frac{\partial^2 l_n(\theta)}{\partial\theta^2}\right)\right\}$$

When the regularity conditions (R0)–(R5) are satisfied and the n observations are i.i.d.,

$$\sqrt{n}(\hat{\theta}-\theta_0) \xrightarrow{D} N\left(0, \frac{1}{I_1(\theta_0)}\right)$$

Statistical Inference for Discrete Data (Cont'd)

Statistical Inference

Null and alternative hypotheses:

$$H_0 : \theta = \theta_0 \quad \text{vs} \quad H_1 : \begin{cases} \theta > \theta_0 \\ \theta < \theta_0 \\ \theta \neq \theta_0 \end{cases}$$

Assumptions: The regularity conditions (R0)–(R5) are satisfied, and the n observations are i.i.d.

Wald test statistic: (under the null hypothesis)

$$\sqrt{I_n(\hat{\theta})}\,(\hat{\theta} - \theta_0) \xrightarrow{D} N(0,1)$$

$$\Leftrightarrow I_n(\hat{\theta})(\hat{\theta} - \theta_0)^2 \xrightarrow{D} \chi^2(df = 1)$$

Statistical Inference for Discrete Data (Cont'd)

Statistical Inference

Score test statistic: (under the null hypothesis)

$$\frac{1}{\sqrt{I_n(\theta_0)}}\left[\frac{\partial l(\theta)}{\partial\theta}\right]_{\theta=\theta_0} \xrightarrow{D} N(0,1) \Leftrightarrow \frac{1}{I_n(\theta_0)}\left[\frac{\partial l(\theta)}{\partial\theta}\right]^2_{\theta=\theta_0} \xrightarrow{D} \chi^2(df=1)$$

Likelihood ratio statistic: (under the null hypothesis)

$$2\left[l(\hat{\theta}) - l(\theta_0)\right] \xrightarrow{D} \chi^2(df=1)$$

Statistical Inference for Discrete Data (Cont'd)

Statistical Inference

(Example) Let $X_1, \dots, X_n \sim iid\ N(\mu, \sigma^2)$

Find the Wald, Score, and Likelihood Ratio test statistics for the hypotheses:

H_0: $\mu = 1$ vs H_1: $\mu \neq 1$,with σ^2 known.

Statistical Inference for Discrete Data (Cont'd)

Discussion

Computational burden:

score statistic ≤ Wald statistic ≤ likelihood-ratio statistic

Test efficiency (power):

score statistic ≤ Wald statistic ≤ likelihood-ratio statistic

Robustness to distributional assumptions (Type I error control):

Wald statistic ≤ score statistic ≈ likelihood-ratio statistic